Draft version June 5, 2026
Typeset using LaTeX twocolumn style in AASTeX7.0.2

A PHANGS-Informed Simulation Suite for the Star-Forming Interstellar Medium: Sampling Design and TIGRESS-NCR Initial Conditions

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ABSTRACT

We present the design of a PHANGS-informed simulation suite for the star-forming interstellar medium (ISM), built on the TIGRESS-NCR framework. Following the philosophy of the CAMELS project in cosmology, we construct an observationally anchored simulation ensemble that spans the range of galactic environments found in nearby galaxies, enabling systematic comparison with observations and simulation-based parameter inference. The PHANGS megatable survey provides aperture-level measurements of gas surface density, stellar surface density, orbital frequency, stellar scale height, and shear across ∼80 nearby galaxies. We use these as an informed prior over the five environmental parameters that enter TIGRESS-NCR-NCR as initial conditions. The sampling method fits a kernel density estimator to the observed joint distribution of these five design fields and maps a scrambled Sobol quasi-random sequence through the marginal quantile functions to generate the simulation design. The Sobol approach enables progressive extension of the suite at any later stage without displacing earlier simulations. Molecular fraction and star formation rate tracers are excluded from the input design and reserved as validation quantities to be compared against emergent simulation outputs. [PLACEHOLDER: summary of simulation results and PHANGS validation to be added]

Keywords: Interstellar medium (847); Star formation (1569); Stellar feedback (1602); Magnetohydrodynamical simulations (1966); Methods: statistical (1361)

1. INTRODUCTION

Star formation in galaxies is a multi-scale, multi-physics process whose regulation has emerged as one of the central problems of galaxy evolution. Across a wide range of galactic environments, observations reveal tight correlations between local gas properties, dynamical equilibrium pressure, and the star formation rate (SFR) surface density (E. C. Ostriker et al. 2010; E. C. Ostriker & R. Shetty 2011; J. Sun et al. 2020; E. C. Ostriker & C.-G. Kim 2022). These correlations are now understood as a signature of self-regulation: energy and momentum injected by massive-star feedback—supernovae, stellar winds, and radiation pressure—maintain the interstellar medium (ISM) in an approximately steady state in which the vertical weight of the gas is supported against gravity by turbulent, thermal, and magnetic pressures. The resulting balance ties the local SFR to the local dynamical state of the disk, and reproduces the observed scaling between $\Sigma_{\rm SFR}$ and the dynamical equilibrium pressure $P_{\rm DE}$ across more than four orders of magnitude in environment (J. Sun et al. 2020, 2023; V. Vijayakumar et al. 2025).

The pressure-regulated, feedback-modulated (PRFM) framework (E. C. Ostriker & C.-G. Kim 2022) provides a theoretical foundation for this self-regulation. In its compact form, vertical equilibrium fixes the midplane total pressure to the weight of the overlying gas, $P_{\rm tot} = \mathcal{W}$, where $\mathcal{W}$ is the integrated weight of the gas column in the combined gravitational potential of the gas, stars, and dark matter. The feedback yield $\Upsilon_{\rm tot} \equiv P_{\rm tot}/\Sigma_{\rm SFR}$ then encapsulates how efficiently young stellar populations convert star formation into the pressure required to support the disk. Equating these two relations yields the prediction

$$\Sigma_{\rm SFR} = \frac{\mathcal{W}}{\Upsilon_{\rm tot}}, \quad (1)$$

in which $\mathcal{W}$ is dictated by the local galactic environment and $\Upsilon_{\rm tot}$ is set by feedback physics. The TIGRESS-NCR simulation framework (C.-G. Kim et al. 2024) resolves the multiphase ISM in local shearing-box magnetohydrodynamic (MHD) simulations with explicit non-equilibrium cooling/heating, UV radiation transfer, cosmic-ray transport, and clustered supernova feedback, and delivers calibrated values of $\Upsilon_{\rm tot}$ and its thermal, turbulent, and magnetic components across metallicities. PRFM has been adopted as the star formation prescription in cosmological-volume simulations (S. Hassan et al. 2024) and isolated-galaxy simulations (S. M. R. Jeffreson et al. 2024), where it provides a physically motivated alternative to empirical Kennicutt–Schmidt-type prescriptions.

The observational counterpart to TIGRESS-NCR is provided by the Physics at High Angular resolution in Nearby GalaxieS (PHANGS) survey (A. K. Leroy et al. 2021; D. R. Weisz et al. 2023), which delivers spatially resolved, multi-wavelength measurements of ~ 80 nearby star-forming disk galaxies. PHANGS aperture catalogs combine high-resolution CO mapping with matched H I, stellar mass, and SFR tracers to produce joint constraints on gas, stars, and star formation at sub-kpc scales (J. Sun et al. 2022, 2023). Together with stellar dynamical modeling of the same galaxies (J. Sun et al. 2020; V. Vijayakumar et al. 2025), the survey defines an empirical envelope of local galactic environments—spanning more than two orders of magnitude in gas and stellar surface density, and the full range of orbital frequencies found in disk-dominated systems—against which any ISM theory must be tested.

To date, however, comparisons between TIGRESS-NCR-type simulations and PHANGS-type observations have relied almost exclusively on a handful of representative simulations selected by hand at the solar neighborhood and a few extreme environments. No systematic survey has yet been carried out that spans the full PHANGS parameter volume with controlled, calibrated simulations of the resolved multiphase ISM. An analogous gap in cosmology—between hand-picked hydrodynamic simulations and the high-dimensional space of cosmological and astrophysical parameters required for inference—was closed by the Cosmology and Astrophysics with MachinE Learning Simulations (CAMELS) project (F. Villaescusa-Navarro et al. 2021). CAMELS demonstrated that a structured simulation ensemble covering parameter space in a controlled way enables emulator construction, simulation-based inference, and systematic comparison with observations, even at the modest individual-simulation fidelity that suite-scale projects can afford. The same logic motivates a CAMELS-style ensemble for the star-forming ISM.

In this paper, the first in a series, we present the design of a PHANGS-informed simulation suite for the star-forming ISM based on the TIGRESS-NCR framework. We use the PHANGS megatable (J. Sun et al. 2022) to construct an observational prior over the five environmental parameters that enter TIGRESS-NCR as initial conditions: the gas surface density $\Sigma_{\rm gas}$, stellar surface density $\Sigma_{\rm star}$, stellar scale height $H_\star$, orbital frequency Ω, and shear parameter q. We construct the simulation design by fitting a kernel density estimator (KDE) to the joint PHANGS distribution of these five fields and mapping a scrambled Sobol quasi-random sequence through its marginal quantile functions. The Sobol construction allows the suite to be extended progressively—doubling the sample size without displacing or re-running earlier simulations—and yields low-discrepancy coverage of the PHANGS environmental volume at every intermediate sample size. The molecular fraction $f_{\rm mol}$ and SFR surface density $\Sigma_{\rm SFR}$ are deliberately excluded from the input design and reserved as validation quantities to be compared against emergent simulation outputs. In this first suite, all physics parameters (metallicity, dust-to-gas ratio, feedback prescription) are held at fiducial solar-neighborhood values; extension to physics-parameter variations and the use of this suite for simulation-based inference (SBI) are the subjects of companion papers CGK: (Paper II, in prep.).

The remainder of this paper is organized as follows. Section 2 describes the PHANGS megatable data, the aperture-level join, and the derivation of the five design fields and validation fields used throughout. CGK: Section 3 presents the sampling design, including the KDE fit, the Sobol construction, and the fairness diagnostics used to assess marginal and joint coverage. Section 4 describes the mapping from PHANGS design fields to TIGRESS-NCR initial conditions and the fiducial physics choices for the first suite. Section 5 presents the resulting simulation design, an initial set of TIGRESS-NCR runs, and validation against PHANGS-observed $f_{\rm mol}$, $\Sigma_{\rm SFR}$, $P_{\rm DE}$, and $\sigma_{\rm eff}$. CGK: Section 6 discusses extensions to physics-parameter variations, the role of this suite in enabling SBI on the resolved ISM, and connections to large-volume cosmological simulations adopting PRFM.

2. PHANGS DATA AND DERIVED QUANTITIES

This section describes the observational dataset that anchors the simulation design. Section 2.1 summarizes the PHANGS megatable survey and the aperture-level join used to construct a single matched table of gas, stellar, and SFR properties. Section 2.2 describes the derivation of the five PRFM design fields from the PHANGS aperture measurements and stellar/dynamical models. Section 2.3 states the quality cuts applied to define the reference set $\mathcal{D}_{\rm ref}$ that drives the sampling design in CGK: Section 3. Table 1 summarizes the design and validation fields with their units and roles.

2.1. *PHANGS Megatable Survey*

We use the publicly released PHANGS megatable products, which compile spatially resolved measurements of cold gas, ionized gas, stellar mass, and SFR for a homogeneous sample of ~ 80 nearby star-forming disk galaxies (A. K. Leroy et al. 2021; J. Sun et al. 2022; D. R. Weisz et al. 2023). The megatables tile each galaxy in matched apertures sampled on a regular grid in galactocentric coordinates, and provide for each aperture aperture photometry of CO(2–1) from PHANGS-ALMA, H I 21 cm from the PHANGS-H I companion programs, stellar mass surface density from environmental masks of Spitzer/IRAC and WISE imaging, and multiple SFR tracers including Hα, far-UV (FUV)+WISE-4 (W4) combinations, and an extinction-corrected Hα+W4 estimator (J. Sun et al. 2022, 2023).

The megatables are released in two complementary aperture definitions: hexagonal apertures, which provide a regular tessellation useful for mapping context and for visual inspection, and two-dimensional Gaussian-weighted apertures, which provide smoothly tapered measurements that mitigate edge effects in the aperture sum and are the canonical aperture for the PHANGS pressure and SFR scaling analyses (J. Sun et al. 2020, 2023). CGK: We adopt the Gaussian-weighted apertures as canonical throughout this work and use the hexagonal apertures only for context plots — please confirm.

For each galaxy, the gas, stellar, and SFR megatables are released as separate FITS tables indexed by galaxy and aperture. We perform an inner join on (galaxy, aperture) across the three tables, retaining only apertures with valid entries in all joined columns. This yields a single matched table of approximately 4.6×10^4 apertures across ~ 80 CGK: galaxies — please verify the final number after quality cuts, which serves as the starting point for the design-field and validation-field selections described below.

2.2. *Derived Quantities*

The five environmental parameters that enter TIGRESS-NCR as initial conditions form the design-field vector

$$\boldsymbol{\theta}_{\rm env} \;=\; (\Sigma_{\rm gas},\, \Sigma_{\rm star},\, H_\star,\, \Omega,\, q)\,. \tag{2}$$

We construct each component as follows.

Gas surface density.—The total neutral gas surface density is the sum of atomic and molecular contributions,

$$\Sigma_{\rm gas} \;=\; \Sigma_{\rm atom} + \Sigma_{\rm mol}, \tag{3}$$

where $\Sigma_{\rm atom}$ is derived from the H I 21 cm aperture photometry and $\Sigma_{\rm mol}$ is derived from the PHANGS-ALMA CO(2–1) aperture photometry assuming the metallicity-dependent CO-to-H_2 conversion factor adopted by J. Sun et al. (2022). Both components include the standard factor of 1.36 to account for helium. The molecular fraction $f_{\rm mol} = \Sigma_{\rm mol}/\Sigma_{\rm gas}$ is computed from the same components and is used only as a validation quantity (see below); it is not part of $\boldsymbol{\theta}_{\rm env}$.

Stellar surface density.—The stellar surface density $\Sigma_{\rm star}$ is taken from the PHANGS stellar mass maps based on Spitzer/IRAC 3.6 μm or WISE-1 3.4 μm imaging, calibrated to mass-to-light ratios using independent color information, following the procedure described in A. K. Leroy et al. (2021) and V. Vijayakumar et al. (2025).

Orbital frequency.—The orbital angular frequency at each aperture is computed from the circular velocity of the host galaxy's rotation curve,

$$\Omega \;=\; \frac{V_{\rm circ}(R)}{R}, \tag{4}$$

where R is the galactocentric radius of the aperture center and $V_{\rm circ}(R)$ is the smooth axisymmetric rotation curve adopted by the PHANGS dynamical modeling pipeline (V. Vijayakumar et al. 2025).

Stellar scale height.—The stellar scale height $H_\star$ is not directly observed and must be inferred from a stellar dynamical model. We adopt the values tabulated by V. Vijayakumar et al. (2025), who fit an axisymmetric stellar mass model constrained by the rotation curve and the observed stellar surface density profile, and report the implied vertical scale height under the assumption that the stellar disk is locally flattened with an axis ratio characteristic of nearby disk galaxies. CGK: Please confirm whether the $H_\star$ used here is the exponential scale height or the sech^2 scale height in the V. Vijayakumar et al. (2025) convention; this matters for the mapping to $z_\star$ in Section 4.

Shear parameter.—The shear parameter quantifies the local logarithmic gradient of the rotation curve and is defined as

$$q \;\equiv\; -\frac{d\ln\Omega}{d\ln R} \;=\; 1 - \frac{d\ln V_{\rm circ}}{d\ln R}. \tag{5}$$

For a flat rotation curve $q = 1$, for solid-body rotation $q = 0$, and for Keplerian rotation $q = 1.5$. We compute q aperture-by-aperture from the numerical derivative of the PHANGS rotation curves at the galactocentric radius of each aperture. Disk-dominated galaxies are physically bounded by $0 < q \leq 1.5$; this bound is imposed at the quality-cut stage (Section 2.3).

2.3. *Quality Cuts and Completeness*

The reference set $\mathcal{D}_{\rm ref}$ used for sampling is defined by validity cuts on the five design fields only:

$$\mathcal{D}_{\rm ref} \;=\; \big\{\, \boldsymbol{\theta}_{\rm env} \;\big|\; \Sigma_{\rm gas},\, \Sigma_{\rm star},\, H_{\star},\, \Omega > 0;\; 0 < q \leq 1.5 \,\big\}. \tag{6}$$

The positivity requirements on $\Sigma_{\rm gas}$, $\Sigma_{\rm star}$, $H_{\star}$, and Ω remove apertures with missing or non-finite values in any of the four quantities. The upper bound $q \leq 1.5$ excludes apertures whose locally fit rotation-curve derivative is super-Keplerian; such cases occur near rotation-curve transitions or in noisy outer-disk apertures and are unphysical for the smooth axisymmetric models that anchor the TIGRESS-NCR initial conditions.

Crucially, we do *not* impose detection cuts on $f_{\rm mol}$, on the H I–CO partition $(\Sigma_{\rm atom}, \Sigma_{\rm mol})$, or on any of the SFR tracers when defining $\mathcal{D}_{\rm ref}$. These quantities have their own, generally narrower, completeness masks: CO is undetected at the lowest gas surface densities, Hα is suppressed in regions of low recent star formation, and W4 saturates in the brightest disks. If validation-field completeness were imposed on the design selection, the PHANGS prior would be biased toward high-detection environments—preferentially CO-bright, Hα-bright, inner-disk apertures—and the resulting simulation suite would under-sample the low-pressure, atomic-dominated outer disks that form a substantial fraction of the total PHANGS disk area. By restricting the selection cuts to the design fields, the prior reflects the full disk-area-weighted distribution of galactic environments, and the validation quantities can be compared against TIGRESS-NCR outputs without introducing a selection bias CGK: (this argument should be verified against the actual PHANGS completeness statistics — please cross-check).

Validation against PHANGS is performed using only those apertures that additionally pass the relevant validation-field completeness masks (detected CO for $f_{\rm mol}$ and $\Sigma_{\rm mol}$, detected SFR tracer for $\Sigma_{\rm SFR}$, and so forth). These masks are applied per-quantity at the validation stage and are not propagated back into $\mathcal{D}_{\rm ref}$. The full set of design and validation fields is summarized in Table 1.

3. SAMPLING DESIGN

This section describes the construction of an observationally informed prior over the environmental parameters that enter TIGRESS-NCR as initial conditions, and the quasi-random design that converts this prior into a concrete set of simulation inputs. The design fields, their identification with PHANGS aperture measurements, and the validation fields reserved for post-simulation comparison are defined in Section 2; the mapping to TIGRESS-NCR initial conditions is detailed in Section 4.

3.1. *Statistical Goal*

The objective of the sampling procedure is to construct an informed prior over the environmental design vector

$$\boldsymbol{\theta}_{\rm env} \;=\; (\Sigma_{\rm gas},\; \Sigma_{\rm star},\; H_{\star},\; \Omega,\; q)\,, \tag{7}$$

conditioned on a target gas surface density band,

$$p\left(\boldsymbol{\theta}_{\rm env} \,\middle|\, \log_{10}\Sigma_{\rm gas} \in \left[\log_{10}\Sigma_{\rm gas}^{\rm tgt} \pm \Delta\right]\right). \tag{8}$$

This prior is *not* an exact resampling of the PHANGS aperture distribution. Exact resampling would inherit the survey's selection effects (e.g. inclination, distance, sensitivity, aperture footprint) and would tile parameter space inefficiently for emulator and simulation-based inference (SBI) training. Instead, the goal is a compact set of simulation inputs that: (i) preserves the dominant observed correlations among the five design fields; (ii) covers the relevant parameter volume sufficiently densely for emulator training and posterior coverage tests; and (iii) excludes parameter combinations that are unphysical or implausible.

We deliberately exclude the molecular gas fraction $f_{\rm mol}$ and the star formation surface density $\Sigma_{\rm SFR}$ from the design vector. Neither is a TIGRESS-NCR input parameter: in the TIGRESS-NCR framework both are emergent outputs of the explicit thermochemistry and self-consistent star formation prescription (C.-G. Kim et al. 2024).CGK: Add a second TIGRESS-NCR reference here if desired (e.g. Kim et al. 2023 ApJS thermochemistry paper); the key 2023ApJS..264...10K is not yet in references.bib and should be verified in ADS before insertion. Requiring valid $f_{\rm mol}$ and $\Sigma_{\rm SFR}$ measurements in the reference set would furthermore bias the prior toward high-completeness environments in which CO and SFR tracers are simultaneously detected, suppressing the very low-$f_{\rm mol}$ and low-$\Sigma_{\rm SFR}$ regions that the simulation suite must cover. These two fields are instead retained as *validation quantities* to be compared against the corresponding emergent distributions from the completed simulation suite (Section 5).

3.2. *Design Fields and Validation Fields*

We partition the relevant PHANGS aperture-level quantities into two disjoint sets: design fields, which serve as TIGRESS-NCR initial conditions and over which the prior is constructed, and validation fields, which are emergent in TIGRESS-NCR and used only for post-simulation comparison.

Table 1. Design and validation fields used in this work.

Field	Symbol	Unit	Role
Design fields $\boldsymbol{\theta}_{\rm env}$ (selection of $\mathcal{D}_{\rm ref}$)			
Gas surface density	$\Sigma_{\rm gas}$	$M_\odot\,{\rm pc}^{-2}$	TIGRESS input
Stellar surface density	$\Sigma_{\rm star}$	$M_\odot\,{\rm pc}^{-2}$	TIGRESS input
Stellar scale height	$H_\star$	pc	TIGRESS input (mapped to $z_\star$)
Orbital frequency	Ω	${\rm km\,s^{-1}\,kpc^{-1}}$	TIGRESS input
Shear parameter	q	dimensionless, $0 < q \le 1.5$	TIGRESS input
Validation fields (not used for selection of $\mathcal{D}_{\rm ref}$)			
Molecular fraction	$f_{\rm mol}$	dimensionless, $0 \le f_{\rm mol} \le 1$	gas-partition validation
Atomic surface density	$\Sigma_{\rm atom}$	$M_\odot\,{\rm pc}^{-2}$	gas-partition validation
Molecular surface density	$\Sigma_{\rm mol}$	$M_\odot\,{\rm pc}^{-2}$	gas-partition validation
SFR surface density	$\Sigma_{\rm SFR}$	$M_\odot\,{\rm kpc^{-2}\,yr^{-1}}$	SFR validation
Dynamical equilibrium pressure	$P_{\rm DE}$	${\rm cm^{-3}\,K}$	PRFM validation
Effective velocity dispersion	$\sigma_{\rm eff}$	${\rm km\,s^{-1}}$	PRFM validation

Note—Design fields are subject only to the validity cuts in Equation 6 and define the reference set $\mathcal{D}_{\rm ref}$ used by the sampling design. Validation fields carry separate per-quantity completeness masks and are compared against emergent TIGRESS-NCR outputs in Section 5.3.

Design fields.—The five-dimensional design vector $\boldsymbol{\theta}_{\rm env}$ (defined in Equation 7) contains the gas surface density $\Sigma_{\rm gas}$, the stellar surface density $\Sigma_{\rm star}$, the stellar scale height $H_\star$, the orbital frequency Ω, and the shear parameter $q \equiv -d\ln\Omega/d\ln R$. These five quantities are sufficient to specify the local shearing-box TIGRESS-NCR initial conditions up to the dark matter midplane density $\rho_{\rm dm}$, which is derived from the rotation curve as described in Section 4.

Validation fields.—The validation set comprises emergent quantities that are diagnostic of the multiphase ISM and star formation, but are not used to specify the simulations:

$$\boldsymbol{\theta}_{\rm env}^{\rm val} = \left(f_{\rm mol},\ \Sigma_{\rm atom},\ \Sigma_{\rm mol},\ \Sigma_{\rm SFR},\ P_{\rm DE},\ \sigma_{\rm eff}\right). \quad (9)$$

Among these, $P_{\rm DE}$ and $\sigma_{\rm eff}$ are derived quantities in the PRFM framework (E. C. Ostriker & C.-G. Kim 2022) that are computed from the design fields via the vertical equilibrium and feedback closure relations; their PHANGS values therefore depend on the same five design fields plus the choice of PRFM effective velocity dispersion model. We use the design-field/validation-field split exclusively to define which distributions are matched in the prior construction (design fields) versus checked in post-simulation comparison (validation fields).

3.3. *Reference Pixel Selection*

The reference dataset $\mathcal{D}_{\rm ref}$ from which the prior is fit is the subset of PHANGS apertures (J. Sun et al. 2022, 2023) that satisfy a target $\Sigma_{\rm gas}$ band and basic physical validity. Specifically, a PHANGS aperture i is included in $\mathcal{D}_{\rm ref}$ if

$$\left|\log_{10}\Sigma_{\rm gas}^{(i)} - \log_{10}\Sigma_{\rm gas}^{\rm tgt}\right| \le \Delta, \quad (10)$$

$$\Sigma_{\rm gas}^{(i)},\ \Sigma_{\rm star}^{(i)},\ H_\star^{(i)},\ \Omega^{(i)} > 0, \quad (11)$$

$$0 < q^{(i)} \le 1.5. \quad (12)$$

The band width Δ defines the local extent of $\mathcal{D}_{\rm ref}$ in $\log_{10}\Sigma_{\rm gas}$ and is chosen to balance two competing requirements: Δ must be wide enough to provide a statistically meaningful sample for the kernel density estimator, yet narrow enough that the conditional joint distribution of the remaining design fields is well approximated as locally stationary in $\log_{10}\Sigma_{\rm gas}$. A fiducial choice $\Delta = 0.2$ dex is adopted in this work and discussed further in Section 5.

The upper bound on the shear parameter, $q \le 1.5$, follows from the definition $q = -d\ln\Omega/d\ln R$: $q = 0$ corresponds to solid-body rotation, $q = 1$ to a flat rotation curve, and $q = 1.5$ to Keplerian rotation around a point mass. Values $q > 1.5$ would correspond to super-Keplerian rotation curves that cannot be produced by a positive enclosed mass distribution and are therefore excluded as unphysical. We similarly exclude $q \le 0$, which corresponds to rotation curves that increase faster than linearly with radius and is not a regime in which the local shearing-box approximation underlying TIGRESS-NCR is expected to apply.

Crucially, validity of $f_{\rm mol}$, $\Sigma_{\rm SFR}$, or any other tracer-specific quantity is *not* required for inclusion in $\mathcal{D}_{\rm ref}$. CGK: This choice maximizes the number of usable reference pixels — particularly in the low-$\Sigma_{\rm gas}$, low-$f_{\rm mol}$ outer-disk regime where CO is undetected and the PHANGS megatable returns upper limits on $\Sigma_{\rm mol}$ — but should be verified against the megatable flag convention to ensure that apertures with all five design fields validly measured are retained.

3.4. *Kernel Density Estimator Prior*

We work throughout in log-transformed design space. Define the five-dimensional log-vector

$$\boldsymbol{w} \;\equiv\; (\log_{10}\Sigma_{\rm gas},\ \log_{10}\Sigma_{\rm star},\ \log_{10}H_\star,\ \log_{10}\Omega,\ \log_{10}q)\,, \tag{13}$$

so that the reference set in log-space is $\{\boldsymbol{w}_i\}_{i=1}^N$ with $N = |\mathcal{D}_{\rm ref}|$. The log transformation is natural because each design field spans more than an order of magnitude across PHANGS and is approximately log-normally distributed within fixed $\Sigma_{\rm gas}$ bands; it also makes the KDE bandwidth selection scale-invariant under multiplicative rescaling of any field.

We fit a Gaussian kernel density estimator to the reference log-vectors,

$$\hat{p}(\boldsymbol{w}) \;=\; \frac{1}{N}\sum_{i=1}^{N} K_{\boldsymbol{H}}\left(\boldsymbol{w}-\boldsymbol{w}_i\right), \tag{14}$$

where $K_{\boldsymbol{H}}$ is the multivariate Gaussian kernel

$$K_{\boldsymbol{H}}(\boldsymbol{u}) \;=\; (2\pi)^{-5/2}\,\det(\boldsymbol{H})^{-1/2}\,\exp\!\left[-\tfrac{1}{2}\,\boldsymbol{u}^\top\boldsymbol{H}^{-1}\boldsymbol{u}\right], \tag{15}$$

and $\boldsymbol{H}$ is a symmetric positive-definite bandwidth matrix. We adopt the multivariate Scott's rule for the full bandwidth matrix,

$$\boldsymbol{H} \;=\; \left(\frac{4}{d+2}\right)^{2/(d+4)} N^{-2/(d+4)}\,\hat{\boldsymbol{\Sigma}}, \tag{16}$$

where $d = 5$ is the number of design dimensions, $N = |\mathcal{D}_{\rm ref}|$ is the reference sample size, and $\hat{\boldsymbol{\Sigma}}$ is the empirical covariance of the N reference log-vectors; for $d = 5$ this gives $\boldsymbol{H} = (4/7)^{2/9}N^{-2/9}\hat{\boldsymbol{\Sigma}}$. Multiplying the full empirical covariance by a scalar factor ensures that $\boldsymbol{H}$ is positive definite and correctly encodes the observed correlations among all five log-fields. In practice we will compare against likelihood-based cross-validation to verify that this bandwidth does not oversmooth the joint distribution. Because all five design fields are strictly positive in $\mathcal{D}_{\rm ref}$ and their upper bounds (most importantly $q \le 1.5$) are handled by the post-mapping rejection step rather than by a boundary-corrected KDE, the KDE in log-space requires no special boundary treatment. This is in contrast to designs that include the molecular fraction $f_{\rm mol} \in [0,1]$, where a logit transform is needed to avoid density leakage outside the physical support.

From $\hat{p}(\boldsymbol{w})$ we draw a large auxiliary KDE sample of size $M \gg N$ (we use $M = 10^5$) and from this sample compute, for each design coordinate $j = 1,\ldots,5$, the marginal cumulative distribution function $F_j(w_j)$ and its inverse, the quantile function

$$Q_j(u) \;=\; F_j^{-1}(u), \qquad u \in (0,1). \tag{17}$$

The quantile functions are tabulated on a dense grid in u and used as look-up tables in the Sobol mapping (Section 3.5). We emphasize that $\hat{p}(\boldsymbol{w})$ retains the full multivariate correlation structure through the off-diagonal entries of $\boldsymbol{H}$, but the marginal quantile functions $\{Q_j\}$ encode only the one-dimensional marginals of $\hat{p}$. The dimension-wise inverse-CDF mapping in Equation 18 therefore produces a sample whose one-dimensional marginal distributions match those of $\hat{p}$ *exactly*, while the joint distribution corresponds to an independence copula applied to these marginals. The joint correlation structure of $\hat{p}$ is thus reproduced only approximately, with fidelity depending on how well an independence copula approximates the true KDE joint. In practice this approximation is adequate when the pairwise correlations among the log design fields are moderate, as is typical within a fixed $\Sigma_{\rm gas}$ band; we assess the quality of the joint match quantitatively in Section 5.

3.5. *Sobol Sequence Design*

Having reduced the prior to its marginal quantile functions on a unit hypercube, we map a quasi-random low-discrepancy sequence through these functions to produce the simulation design. We adopt a scrambled Sobol sequence (I. M. Sobol' 1967; A. B. Owen 1998) rather than a pseudo-random or Latin Hypercube Sampling (LHS) design for two reasons. First, for any finite sample size n, Sobol sequences achieve a star-discrepancy of $O(n^{-1}(\log_{10}n)^d)$ in d dimensions, asymptotically smaller than the $O(n^{-1/2})$ scaling of pseudo-random samples and uniformly better than LHS in projections onto coordinate subspaces. Second, and more importantly for an incrementally grown simulation suite, the Sobol construction has a nested progressive property: the first 2^k points of the sequence form a balanced design at all k, so doubling the sample size from 2^k to 2^{k+1} adds new design points without displacing any of the original ones. LHS, by contrast, must in general be re-stratified — and therefore re-drawn from scratch — whenever the sample size changes.

Sampling algorithm.—Given a target sample size n, the design is generated as follows.

1. Draw a scrambled Sobol sequence $\{\boldsymbol{u}_k\}_{k=1}^{n} \subset [0,1]^5$ using `scipy.stats.qmc.Sobol(d=5, scramble=True)` with a fixed random seed for reproducibility.

2. Map each Sobol point through the marginal quantile functions of Equation 17 to obtain log-space design points,

$$w_{k,j} \;=\; Q_j\left(u_{k,j}\right), \qquad j = 1, \ldots, 5. \tag{18}$$

3. Convert to physical-space design points,

$$\theta_{k,j} \;=\; 10^{\,w_{k,j}}, \tag{19}$$

yielding $\boldsymbol{\theta}_k = (\Sigma_{\rm gas}, \Sigma_{\rm star}, H_\star, \Omega, q)_k$.

4. For each design point with $q_k > 1.5$, draw the next un-used Sobol point from the sequence and repeat steps 2–3 until an accepted point is obtained. Let $n_{\rm extra}$ denote the total number of rejected-and-replaced draws. The accepted set of n points is formed from the first $n + n_{\rm extra}$ Sobol indices, with the $n_{\rm extra}$ rejected indices recorded so that future extensions begin from index $n + n_{\rm extra} + 1$. This bookkeeping preserves the progressive nesting property (Section 3.6).

The shear bound is the only post-mapping rejection criterion; positivity is automatically guaranteed by the log-space construction. Because the q marginal in $\mathcal{D}_{\rm ref}$ is already upper-truncated by Equation 12, the rejection fraction $n_{\rm extra}/n$ is expected to be small; we report the realized value in Section 5.

Sample sizes that are powers of two, $n = 2^m$ with m a non-negative integer, are preferred because the Sobol sequence's discrepancy bounds are sharp at these sizes and the construction is balanced in projections to dyadic subintervals. Arbitrary n is permitted at the cost of a mild loss of projection uniformity.

3.6. *Progressive Extension of the Suite*

A central operational advantage of the Sobol-based design is that the suite is extensible at any later stage without re-drawing the existing simulations. Let $S(n)$ denote the set of accepted design points produced by the algorithm of Section 3.5 at sample size n. The underlying Sobol sequence has the nesting property: the first 2^k sequence points are a subset of the first 2^{k+1} sequence points. Provided the same random seed and the same physical rejection criterion ($q > 1.5$) are applied at every stage, the accepted-point sets inherit this property,

$$S(2^k) \;\subset\; S(2^{k+1}) \;\subset\; S(2^{k+2}) \;\subset\; \cdots, \tag{20}$$

so that any simulation completed at stage k is automatically a member of the design at every subsequent stage. In practice we generate $n + n_{\rm extra}$ Sobol points at each stage, accept the first n points that pass the rejection criterion, and record the consumed Sobol indices so that future extensions begin from the next un-used point in the same sequence. The first generation of the suite targets $n \in \{64, 128, 256\}$ ($m \in \{6, 7, 8\}$), corresponding to a sequence of doublings; this progression is set by the available computational allocation rather than any statistical threshold, and the discrepancy of the design improves monotonically along the sequence.

At each stage, the new 2^k points fill the largest remaining gaps in the previous-stage coverage of the unit hypercube and, by the inverse-CDF mapping of Equation 18, the largest remaining gaps in log-design space weighted by the KDE prior. This contrasts sharply with LHS-based designs, which must be re-stratified to add new points and therefore either displace the existing design or accept a strictly sub-optimal stratification on the enlarged design.

This progressive structure is closely analogous to the staged extension strategy adopted by the CAMELS project (F. Villaescusa-Navarro et al. 2021), in which cosmological hydrodynamic simulation suites have been incrementally enlarged as new scientific questions are identified. The Sobol nesting provides for the PHANGS-informed TIGRESS-NCR suite the same operational property that CAMELS achieves through its explicit batch design: completed simulations are never invalidated by the addition of new ones.

3.7. *Block-Structured Extension to Physics Parameters*

The first-generation suite holds all physics parameters fixed at fiducial solar-neighborhood values (Section 4). Subsequent generations will vary uncertain physics parameters such as the gas-phase metallicity Z'_g, the dust-to-gas ratio Z'_d, the cosmic-ray ionization rate $\xi_{\rm cr}$, the assumed initial mass function, and parameters of the supernova feedback prescription. We collect these into a $d_{\rm phys}$-dimensional physics design vector $\boldsymbol{\theta}_{\rm phys}$.

We adopt a block-structured Sobol design over $(\boldsymbol{\theta}_{\rm env}, \boldsymbol{\theta}_{\rm phys})$, defined as follows. Let

$$\{\boldsymbol{u}_k^{\rm env}\}_{k=1}^{n_{\rm env}} \;\subset\; [0,1]^5, \qquad \left\{\boldsymbol{u}_\ell^{\rm phys}\right\}_{\ell=1}^{n_{\rm phys}} \;\subset\; [0,1]^{d_{\rm phys}} \tag{21}$$

be two independent scrambled Sobol sequences, one over the environmental block and one over the physics block.

The combined design is constructed by index-aligned pairing: simulation k uses environmental point $\boldsymbol{u}_k^{\mathrm{env}}$ and physics point $\boldsymbol{u}_k^{\mathrm{phys}}$, so the total number of simulations equals $\min(n_{\mathrm{env}}, n_{\mathrm{phys}})$. Each block is mapped through its respective prior: $\boldsymbol{u}_k^{\mathrm{env}}$ through the KDE-based quantile functions of Equation 18, and $\boldsymbol{u}_k^{\mathrm{phys}}$ through prior quantile functions appropriate to each physics parameter (e.g. log-uniform in Z_g' over a chosen range). When the two blocks differ in size, the smaller block is extended by drawing additional Sobol points before pairing.

The block construction has two advantages. First, the two blocks may be extended independently: doubling n_{env} at fixed n_{phys} yields a denser environmental sampling at the original set of physics points, while doubling n_{phys} at fixed n_{env} adds physics variations to the existing environmental design. Both extensions inherit the nesting property of Equation 20 within their respective blocks. Second, the two priors are kept manifestly independent, which is desirable when the physics-parameter prior is itself uncertain or under active revision: an update to $p(\boldsymbol{\theta}_{\mathrm{phys}})$ does not require re-drawing the environmental design.

This block structure is analogous to the separation between "cosmological" and "astrophysical" parameter suites in the CAMELS program (F. Villaescusa-Navarro et al. 2021), which treats the two parameter groups as independent design axes that can be extended along either axis without re-running the other.

Option A: pre-allocated joint design.—If the full set of varied physics parameters and their priors are known *a priori*, an alternative is to draw a single $(5 + d_{\mathrm{phys}})$-dimensional Sobol sequence over the joint space and map each block through its respective prior. This yields the optimal joint discrepancy at all sample sizes and preserves the nesting property of Equation 20 in the joint space, but at the cost of fixing d_{phys} from the outset: adding a new physics parameter later requires re-drawing the design. We therefore prefer the block-structured construction for the staged suite considered here, and reserve the pre-allocated joint design for a future generation in which the physics-parameter list is finalized.

3.8. *Expanded Prior for Rare Environments*

The KDE prior of Section 3.4 faithfully reproduces the PHANGS design-field distribution within the target Σ_{gas} band, but by construction it under-samples regions of parameter space that are under-represented in PHANGS. Such regions include, for example, very low orbital frequencies Ω characteristic of low-surface-brightness or outer-disk environments, large stellar scale heights $H_\star$ characteristic of thickened or perturbed disks, and shear parameters q near the rising-rotation-curve ($q \to 0$) or Keplerian ($q \to 1.5$) extremes. For emulator and SBI training it is desirable to extend coverage into these tails to guard against extrapolation in posterior inference.

We implement such extension in two complementary ways. (i) Broadening the KDE bandwidth, $\boldsymbol{H} \to s\,\boldsymbol{H}$ with $s > 1$, inflates the prior in all directions while preserving the empirical mean and correlation structure; this is sufficient for modest extrapolation. (ii) Directional extension of the acceptance bounds in selected log-space directions, implemented by augmenting the quantile functions $Q_j(u)$ at the tails (i.e. extending Q_j for u near 0 or 1 beyond the support of $\mathcal{D}_{\mathrm{ref}}$), supports targeted extension into rare regimes such as low-Ω environments. Expanded-prior simulations are flagged at draw time and analyzed separately from the PHANGS-prior simulations in the fairness diagnostics (Section 3.9) and downstream emulator validation. CGK: The precise broadening factor s and the directional extensions applied in the first-generation suite will be reported in Section 5; the choice of these is closely tied to which PHANGS-poor environments the inference program ultimately targets.

3.9. *Sampling Fairness Diagnostic*

We define a quantitative diagnostic of how faithfully the simulation design reproduces the PHANGS design-field distribution within the target Σ_{gas} band. For each design field $a \in \boldsymbol{\theta}_{\mathrm{env}}$, let $\hat{q}_a^{\mathrm{sample}}(\tau)$ and $\hat{q}_a^{\mathrm{ref}}(\tau)$ denote the empirical τ-th quantile of $\log_{10\,10} a$ in the simulation design and in $\mathcal{D}_{\mathrm{ref}}$, respectively (we use τ for quantile probability levels to distinguish from the KDE marginal quantile functions Q_j of Equation 17). The per-field quantile mismatch is

$$\varepsilon_a = \max_{\tau \in \{0.1,\,0.2,\,\ldots,\,0.9\}} \left| \hat{q}_a^{\mathrm{sample}}(\tau) - \hat{q}_a^{\mathrm{ref}}(\tau) \right|, \qquad (22)$$

in units of dex, and the overall design fairness is the worst-field mismatch,

$$\varepsilon_{\mathrm{env}} = \max_{a \in \boldsymbol{\theta}_{\mathrm{env}}} \varepsilon_a. \qquad (23)$$

A design is considered *fair* with respect to $\mathcal{D}_{\mathrm{ref}}$ if $\varepsilon_{\mathrm{env}} \le \varepsilon_{\mathrm{tol}}$, where we adopt $\varepsilon_{\mathrm{tol}} = 0.10$–$0.20\,$dex as a fiducial tolerance. This range is motivated by the typical aperture-level measurement uncertainty in the underlying PHANGS design fields, which is itself of order $\sim 0.1\,$dex for the surface densities and somewhat larger for $H_\star$; a design that matches the reference marginal quantiles to better than the measurement uncertainty is not meaningfully distinguishable from the reference within $\mathcal{D}_{\mathrm{ref}}$. CGK: The 0.10–$0.20\,$dex tolerance is a working choice; a quantitative justification from the PHANGS measurement-uncertainty budget should be added when those numbers are tabulated.

The diagnostic in Equation 23 applies exclusively to the five *design* fields. The validation fields $f_{\rm mol}$, $\Sigma_{\rm atom}$, $\Sigma_{\rm mol}$, $\Sigma_{\rm SFR}$, $P_{\rm DE}$, and $\sigma_{\rm eff}$ are not used to define the prior, and they cannot be matched at design time because they are not TIGRESS-NCR inputs. Instead, the simulation outputs for the validation fields will be compared against the corresponding PHANGS distributions in Section 5 using analogous quantile-mismatch metrics, providing a post-hoc verification that the simulated ISM populates the same emergent regime as the observed one. A successful program is one in which $\varepsilon_{\rm env} \leq \varepsilon_{\rm tol}$ by design and the corresponding validation-field mismatches are also small without having been tuned.

4. TIGRESS-NCR PARAMETER MAPPING

4.1. *TIGRESS-NCR Input Parameters*

[SKELETON] The TIGRESS-NCR framework (C.-G. Kim et al. 2024) solves the equations of ideal MHD with self-gravity and explicit thermochemistry in a local shearing-box geometry. The initial conditions are specified by the following parameters:

- $\Sigma_{\rm gas}$: total gas surface density
- $\Sigma_{\rm star}$: stellar surface density
- $z_\star$: stellar scale height
- Ω: orbital frequency
- $\rho_{\rm dm}$: dark matter volume density at the midplane
- q: shear parameter $q \equiv -d\log_{10}\Omega/d\log_{10}R$

4.2. *Mapping from PHANGS to TIGRESS-NCR*

[SKELETON] The PHANGS design fields map to TIGRESS-NCR inputs as follows:

$$(\Sigma_{\rm gas}, \Sigma_{\rm star}, H_\star, \Omega, q)_{\rm PHANGS} \longrightarrow (\Sigma_{\rm gas}, \Sigma_{\rm star}, z_\star, \Omega, \rho_{\rm dm}, q)_{\rm TIGRESS} \quad (24)$$

The stellar scale height $H_\star$ is related to $z_\star$ by ... [MAPPING TO BE DOCUMENTED].

The dark matter density $\rho_{\rm dm}$ is inferred from the circular velocity curve as ... [MAPPING TO BE DOCUMENTED].

4.3. *Fiducial Physics Parameters*

The first simulation suite holds all physics parameters at fiducial solar-neighborhood values: metallicity $Z'_g = 1$, dust-to-gas ratio $Z'_d = 1$, and standard feedback and population-synthesis assumptions. Extension to physics parameter variations is described in Section 6.

5. RESULTS

5.1. *Sampling Diagnostics*

Figure 1 shows the marginal distributions of the five design fields for the KDE-Sobol samples at $n = 64$ and $n = 128$, compared to the PHANGS reference set. The fairness metric $\epsilon_{\rm env}$ (**??**) is reported for each sample size.

Figure 1. Marginal distributions of the five environmental design fields ($\Sigma_{\rm gas}$, $\Sigma_{\rm star}$, $H_\star$, Ω, q) for the KDE-Sobol samples ($n = 64$: blue; $n = 128$: orange) compared to the PHANGS reference set (gray). Quantile fairness ϵ is indicated for each field. *[PLACEHOLDER — figure to be updated with Sobol-based results]*

5.2. *Simulated ISM Properties*

[PLACEHOLDER] This section will present the TIGRESS-NCR simulation results for the first suite of n runs with PHANGS-informed environmental parameters at fiducial metallicity $Z'_g = 1$.

Expected outputs to be reported:

- Time-averaged $\Sigma_{\rm SFR}$, $f_{\rm mol}$, $P_{\rm DE}$, $\sigma_{\rm eff}$ for each simulation
- ISM phase structure and midplane pressure as functions of $\Sigma_{\rm gas}$, $\Sigma_{\rm star}$, Ω
- PRFM feedback yield calibration from the suite

5.3. *Validation Against PHANGS*

[PLACEHOLDER] This section will compare the emergent simulation properties against the observed PHANGS distributions.

Figure 2. Comparison of emergent $\Sigma_{\rm SFR}$ and $f_{\rm mol}$ from the TIGRESS-NCR suite against the observed PHANGS distributions at matched $\Sigma_{\rm gas}$. *[PLACEHOLDER — results pending]*

6. DISCUSSION AND FUTURE DIRECTIONS

The simulation design presented in this paper is the first stage of a broader program: an observationally anchored, progressively extensible ensemble of resolved multiphase ISM simulations that supports systematic comparison with PHANGS and statistical inference of ISM physical parameters. In this section we recapitulate the sampling design (Section 6.1), outline how the block-structured Sobol construction will be extended to physics parameters (Section 6.2), describe the synthetic observation pipeline that will close the loop with resolved PHANGS maps (Section 6.3), discuss summary statistics beyond one-point aperture quantities

Table 2. PHANGS design field to TIGRESS-NCR input mapping

PHANGS field	Unit	TIGRESS parameter	Notes
$\Sigma_{\rm gas}$	$M_\odot$ pc^{-2}	$\Sigma_{\rm gas}$	direct
$\Sigma_{\rm star}$	$M_\odot$ pc^{-2}	$\Sigma_{\rm star}$	direct
$H_\star$	pc	$z_\star$	[conversion TBD]
Ω	km/s/kpc	Ω	direct
q	—	q	direct, $0 < q \leq 1.5$
—	—	$\rho_{\rm dm}$	derived from rotation curve

(Section 6.4), and frame the connection to the inference framework developed in the companion paper (Section 6.5).

6.1. *Summary of the Sampling Design*

We have presented a PHANGS-informed simulation design for the TIGRESS-NCR framework. Five environmental parameters—the gas surface density $\Sigma_{\rm gas}$, stellar surface density $\Sigma_{\rm star}$, stellar scale height $H_\star$, orbital frequency Ω, and shear parameter q—define the design vector $\boldsymbol{\theta}_{\rm env}$ and enter TIGRESS-NCR as initial and boundary conditions. The sampling proceeds in two steps: a kernel density estimator (KDE) is fit to the joint PHANGS aperture distribution of $\boldsymbol{\theta}_{\rm env}$ and used to define the marginal quantile functions, and a scrambled Sobol quasi-random sequence is mapped through these quantile functions to generate the simulation design.

Two features of this construction are essential for the program as a whole. First, because the Sobol sequence is deterministic and hierarchical, the suite can be extended progressively: doubling the sample size adds new low-discrepancy points without displacing or re-running any earlier simulation. Second, the molecular fraction $f_{\rm mol}$ and SFR surface density $\Sigma_{\rm SFR}$ are deliberately excluded from the input design and reserved as validation quantities, to be compared against emergent TIGRESS-NCR outputs. CGK: This separation of design fields from validation fields is what allows the suite to be used both as a calibration set (for PRFM yields) and as a prior for simulation-based inference; flag for review that this framing is consistent with the abstract.

6.2. *Extension to Physics Parameter Variations*

The first suite holds the physics parameters at fiducial solar-neighborhood values: solar metallicity ($Z'_g = 1$), solar dust-to-gas ratio ($Z'_d = 1$), a standard initial mass function (IMF), and the fiducial TIGRESS-NCR-NCR feedback prescription. This choice isolates the environmental dependence of ISM properties from physics variations, but it also means that the inferred posteriors from this suite are conditioned on these fiducial physics assumptions. The natural next step is to relax this conditioning by introducing an independent block of simulations that varies a vector of physics parameters $\boldsymbol{\theta}_{\rm phys}$.

We anticipate four parameters as the principal targets of this extension. First, the gas-phase metallicity $Z'_g \in [0.1, 3]$ in solar units, motivated by the TIGRESS-NCR-NCR metallicity suite of C.-G. Kim et al. (2024), who showed that $\Sigma_{\rm SFR}$ scales weakly with metallicity at fixed environment, approximately as $\Sigma_{\rm SFR} \propto Z_g'^{0.3}$. CGK: Verify the exponent and the sign; I am recalling this from the K24 abstract but please confirm against the paper before this text is finalized. Second, the dust-to-gas ratio Z'_d, which is strongly but not perfectly correlated with Z'_g in observed galaxies and which controls photoelectric heating, H_2 formation, and radiation transfer independently of gas-phase line cooling. Third, the upper-mass cutoff of the IMF, which sets the ionizing photon budget and the supernova rate per unit star formation. Fourth, an overall feedback yield normalization that re-weights the thermal versus turbulent partitioning of $\Upsilon_{\rm tot}$ relative to its calibrated TIGRESS-NCR-NCR value; this last parameter is included not because we expect the calibration to be wrong, but to allow the inference framework to detect inconsistencies between the simulated and observed yield decompositions.

The block-structured Sobol construction makes this extension natural. Block 1 is the present environmental suite, a Sobol design of dimension $d_{\rm env} = 5$ over $\boldsymbol{\theta}_{\rm env}$ at fiducial physics. Block 2 will be a Sobol design of dimension $d_{\rm phys}$ over $\boldsymbol{\theta}_{\rm phys}$, run at a representative subset of environmental anchor points selected from Block 1. The joint suite supports inference over the concatenated parameter vector $(\boldsymbol{\theta}_{\rm env}, \boldsymbol{\theta}_{\rm phys})$ by combining the marginal information from each block under an assumed factorization of the prior. This mirrors the multi-suite philosophy of the CAMELS project (F. Villaescusa-Navarro et al. 2021), in which one-parameter variation suites, Latin-hypercube suites, and extension suites jointly cover the cosmology and astrophysics parameter space at controlled cost. CGK: Verify that this is a faithful characterization of the CAMELS suite structure; the high-level analogy is correct but the suite labels and their exact roles should be checked.

6.3. *Synthetic Observation Pipeline*

A direct comparison between TIGRESS-NCR and PHANGS at the level of integrated aperture quantities, as performed in Section 5, is a necessary but not sufficient test of the simulation suite. The same aperture-averaged $\Sigma_{\rm SFR}$, $f_{\rm mol}$, $P_{\rm DE}$, and $\sigma_{\rm eff}$ can be produced by simulations that differ substantially in the spatial structure of their gas and young stellar populations. To test the simulations at the level of resolved morphology, and to construct simulated observables that can be ingested by the inference framework of the companion paper, we plan to develop a forward-modeling pipeline that converts TIGRESS-NCR snapshots into mock multi-wavelength maps matched to the PHANGS resolution and noise properties.

The pipeline will produce five primary tracers. (i) Mid-infrared dust emission, modeled from the TIGRESS-NCR dust column and ambient radiation field to produce synthetic JWST/MIRI and *Spitzer*/MIPS $24\,\mu$m maps. (ii) Ionized-gas Hα emission, computed from the H II region catalog of each TIGRESS-NCR snapshot with extinction applied self-consistently from the simulated dust column. (iii) Far-ultraviolet continuum, computed from the young stellar population with the same extinction treatment. (iv) H I 21 cm column, derived from the neutral atomic phase of TIGRESS-NCR-NCR. (v) CO line emission, computed either from a calibrated CO-to-H_2 conversion as a function of metallicity and radiation field, or from direct radiative transfer post-processing of the H_2 phase. CGK: Decide which CO route to adopt; both have a literature trail and the choice will need to be justified explicitly. Each tracer will then be convolved with the appropriate beam, projected into the observational aperture grid, and resampled to the PHANGS sensitivity and noise level.

This synthetic observation pipeline serves three purposes. It allows direct comparison of TIGRESS-NCR outputs to observed PHANGS maps in the observed plane, removing the modeling assumptions implicit in converting observed maps to physical surface densities. It produces the maps required as inputs to field-level inference (Section 6.5). And it provides a self-consistent training set for emulators that predict observable maps directly from $\boldsymbol{\theta}_{\rm env}$, bypassing the intermediate step of physical-parameter estimation.

6.4. *Beyond One-Point Statistics*

The validation and inference targets used in this paper are one-point aperture summaries: the mean $\Sigma_{\rm SFR}$, $f_{\rm mol}$, $P_{\rm DE}$, $\sigma_{\rm eff}$, and feedback yield $\Upsilon_{\rm tot}$ within a fixed aperture. These quantities are informative about the pressure balance and the mean efficiency of feedback, but they discard the spatial coherence of star formation and gas distribution that is plausibly the most discriminating signature between feedback models with similar globally averaged behavior.

A natural first extension is to two-point statistics: the angular power spectrum or two-point correlation function of $\Sigma_{\rm gas}$, $\Sigma_{\rm SFR}$, and $P_{\rm DE}$ maps within each aperture or across each simulated patch. Two-point statistics constrain the spatial coherence length of star formation, the slope of the turbulent cascade in the cold gas, and the relative phasing of gas and SFR tracers. CGK: Mention specific turbulent-cascade citations here if desired; I have left this generic because I do not have a high-confidence ADS key in hand for the most appropriate references.

A more powerful and increasingly mature class of statistics for non-Gaussian fields is the wavelet scattering transform (**?**). Wavelet scattering coefficients are translation-equivariant, rotationally invariant, multi-scale descriptors that capture the morphological content of a field at a hierarchy of spatial scales and orientation interactions. They are strictly more informative than two-point statistics for fields with non-Gaussian structure and have been used as field-level summary statistics in ISM, CMB, and galaxy-morphology inference contexts (**?**). CGK: Suggested references to fill in: Allys+19, Cheng+20, Régaldo-Saint Blancard+20/+23; please verify ADS keys and substitute.

In the long term, the goal is field-level inference: training a neural posterior estimator directly on simulated maps, with wavelet scattering coefficients (or learned summary statistics) serving as the intermediate representation. This program collapses the distinction between forward modeling and inference: each TIGRESS-NCR snapshot produces, via the pipeline of Section 6.3, the same observables that PHANGS measures, and the neural estimator learns the posterior over $\boldsymbol{\theta}_{\rm env}$ (and eventually $\boldsymbol{\theta}_{\rm phys}$) directly from the resolved map. The companion paper takes the first step toward this goal using aperture-level summary statistics; the present suite is sized and structured so that the same simulations can be reused as the training set as field-level summaries are added.

6.5. *Connection to the Companion Inference Paper*

The simulation suite presented here is designed from the outset to serve as the numerical prior and simulator for the simulation-based inference (SBI) framework developed in the companion paper CGK: (?; Paper II, in preparation). The sampling design of Section 3, mapped through the KDE of the PHANGS joint distribution, defines the implicit prior $p(\boldsymbol{\theta}_{\rm env})$ under which the inference is performed. Summary statistics extracted from each TIGRESS-NCR snapshot—$\Sigma_{\rm SFR}$, $f_{\rm mol}$, $P_{\rm DE}$, $\sigma_{\rm eff}$, and the decomposed feedback yields $\Upsilon_{\rm th}$ and $\Upsilon_{\rm turb+mag}$—constitute the simulated observables that are matched against PHANGS aperture measurements. The progressive extensibility of the Sobol design ensures that the training set can be enlarged without invalidating any inference performed on earlier subsets, and the reservation of $f_{\rm mol}$ and $\Sigma_{\rm SFR}$ as validation quantities means that the inferred posteriors can be cross-checked against quantities that did not enter the design. The present paper completes the construction of the simulator; the companion paper applies it.

ACKNOWLEDGMENTS

The authors acknowledge the use of Claude Code (Anthropic) as an AI-assisted editing tool during the preparation of this manuscript. All scientific ideas, physical interpretations, mathematical formulations, and intellectual content are original contributions of the authors.

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